EARTH AND PLANETARY SCIENCE LETTERS 17 (1972) 159–168. NORTH-HOLLAND PUBLISHING COMPANY

PALEOMAGNETISM OF MIDWAY ATOLL LAVAS AND NORTHWARD MOVEMENT OF THE PACIFIC PLATE*

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Received 4 August 1972
Revised version received 21 September 1972

Two deep drill holes through the reef limestones of Midway Atoll penetrated 120 m and 19 m of basaltic lavas that were dated by the K–Ar method at 18 my. Inclinations of natural remanent magnetization have been measured in 173 specimens cut from 57 core samples from 13 of the lava flows. The mean paleomagnetic inclination is $27.6^\circ \pm 6.8^\circ$, corresponding to a paleolatitude of $14.7^\circ \pm 4.2^\circ$. The present latitude of Midway is 28°, suggesting a northward component of motion of the Pacific plate of approximately 13° or 1400 km in the last 18 my. The paleolatitude of Midway is thus not significantly different from the present latitude (19°) of the active volcanic island of Hawaii. The paleomagnetic data from the Midway basalts thus support the hypothesis of Wilson and Morgan that volcanic heat sources are fixed with respect to the Earth's mantle below the asthenosphere and their apparent migration with time is due to plate motion.

1. Introduction

The Pacific plate is one of the largest of the Earth's crustal plates and, judging by the spacing and configuration of linear marine magnetic anomalies (see e.g., Hays and Pitman [1]), its motion has been rapid and complex relative to other crustal plates for at least the past 80 my. Although the apparent rate of movement of the Pacific plate is great enough that paleomagnetic poles determined from rocks older than about 10 my should show systematic displacements, such data are very scarce because of the lack of exposed rock in this oceanic area. A preliminary polar wandering curve for the Pacific plate from the Cretaceous to the present has been published by Francheteau et al. [2]. This curve was derived mainly from a set of paleomagnetic poles obtained from analyses of magnetic anomalies over groups of Pacific seamounts. These authors also included an estimate of the latitude of formation of the Great Magnetic Bight and a paleomagnetic determination on two lavas from a drill hole on Midway Atoll, both by Vine [3]. In spite of its preliminary nature, a striking and unmistakable feature of the polar wandering curve of Francheteau et al. is the indication of systematic northward displacement of the Pacific plate, amounting to some 30° since mid-Cretaceous time.

Other paleomagnetic data for the Pacific plate have been obtained in extensive studies of lavas of the Hawaiian chain. These rocks, however, range in age from Late Pleistocene and Holocene [4–6] to Pliocene; the oldest are the 4.5 – 5.6-my-old Napali Formation on Kauai [7, 8]. The paleomagnetic poles derived from these lavas would therefore not be expected to be significantly displaced as a result of Pacific plate motion.

The history of movement of the Pacific plate is especially important with respect to two recent theories concerning the origin of aseismic volcanic island chains and submarine ridges. The basic idea was advanced by Wilson [9, 10], who noted that ages of rocks along these chains and ridges increase progressively away from the sites of active volcanism and that the directions of increasing age are mostly the same for single crustal plates. Wilson concluded that the loci of active volcanism remain essentially fixed with respect to the mantle and

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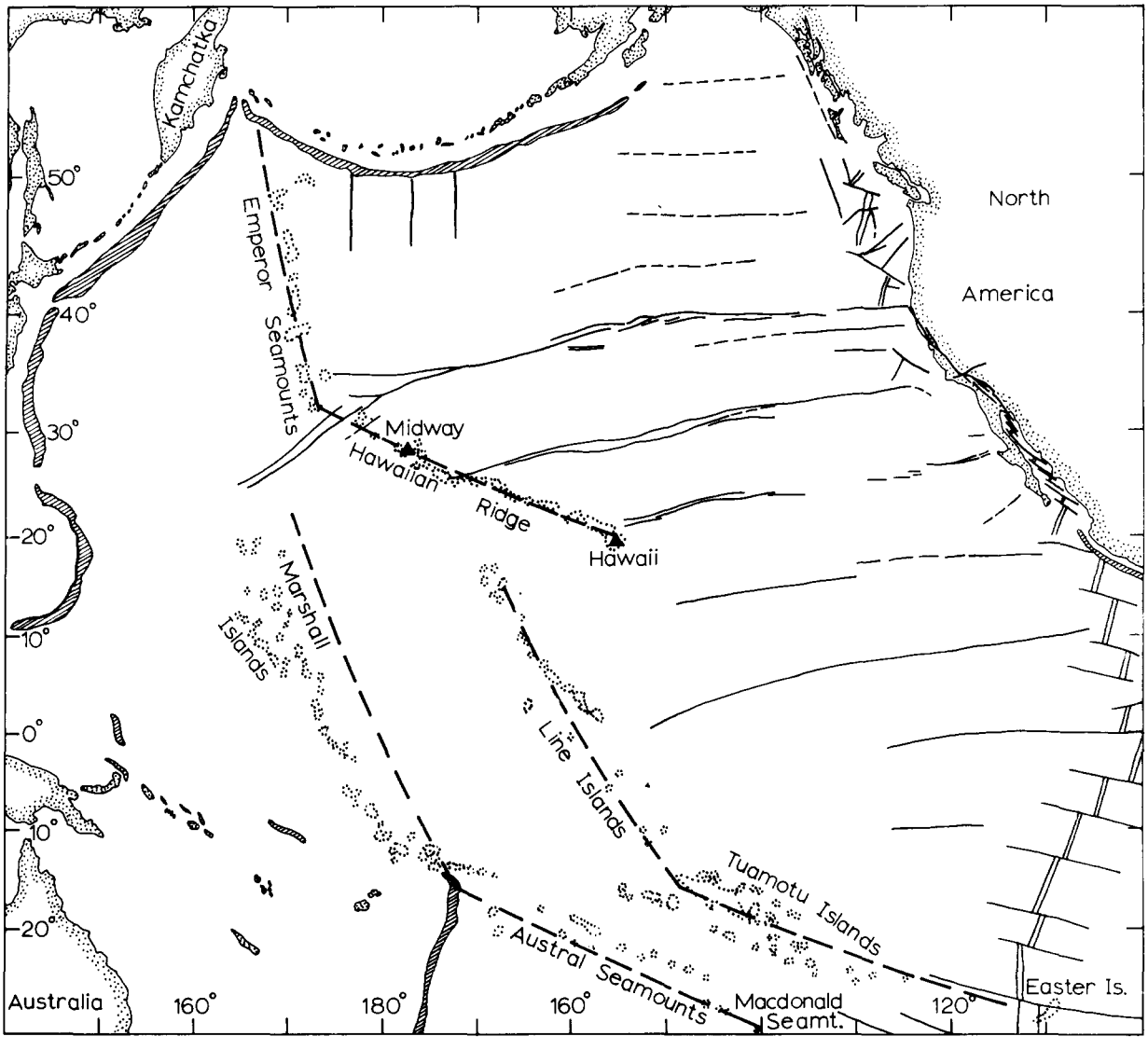


Fig. 1. Sketch map of the Pacific plate, showing aseismic volcanic chains (dotted outlines), marginal trenches (hatched), fracture zones (solid and dashed lines), and spreading centers (double lines). Heavy dashed lines are melting-spot traces proposed by Morgan [12]. Fracture zones and spreading centers are from Hayes and Pitman [1] and Atwater and Menard [16].

that the motion of crustal plates away from median ridges carries the volcanic structures away from their sources.

A modification of Wilson's hypothesis has been developed by Morgan [11, 12], in which the loci of present volcanism, or melting spots, are considered to be nearly fixed with respect to the mantle below the asthenosphere. These melting spots, of which there are about 20, are the surface manifestations of the tops of localized upward convection currents, or "plumes", originating in the lower mantle. The lateral motion in the asthenosphere (upper mantle) radially away from the plumes is supposed to provide the tractive force for plate motion, and the downward counterflow is diffused throughout the remaining bulk of the mantle. According to Morgan [12], the lifetimes of some of these plumes may be very great, at least as old as the initiation of the Atlantic Ocean rifting, approximately 180

my. The hypothesis of fixed melting spots is very important because they potentially provide a fixed frame of reference to which paleomagnetic pole positions and crustal plate motions may be referred. Whether or not true polar wandering has occurred during Cretaceous and Tertiary time is still an open question [13, 14], and the existence of fixed coordinates would allow apparent polar wandering due to plate motion to be distinguished from movements of the pole with respect to the mantle below the asthenosphere.

Another modification of Wilson's hypothesis has been published by McDougall [15], who summarized the radiometric age data for the Hawaiian Island chain which show clearly the progression of volcanism southeastward to the presently active island of Hawaii. Because the rate of migration of the locus of active volcanism is roughly twice the half-rate of spreading at the East Pacific Rise, McDougall concluded that the zone in the upper mantle of return convective flow toward the East Pacific Rise has essentially the same thickness as the Pacific crustal plate and hence moves with the same velocity in the opposite direction, carrying with it the ultimate source of the magmatic activity.

The Pacific plate is especially favorable for testing the fixed-melting-spot hypothesis because it has apparently had a large latitudinal component of absolute motion during Cretaceous and Tertiary time, as implied by the paleomagnetic polar wandering curve of Francheteau et al. [2]. This motion is further illustrated in fig. 1, which shows the present locations of three major melting spots proposed by Morgan [12] (Hawaii, Macdonald Seamount, and apparently the East Pacific Rise west of Easter Island), and the chains of seamounts and islands that presumably resulted from the motion of the Pacific plate over them. If, for example, the Hawaiian melting spot has remained fixed, then all of the volcanic rocks along the Hawaiian–Emperor chain would have been formed at the present latitude of Hawaii [17]. If the geomagnetic field has been predominantly a geocentric axial dipole for a sufficiently long time, i.e, if true polar wandering has been insignificant, then the paleomagnetic inclinations in these rocks should all be the same as the axial dipole inclination at Hawaii. Note that this paleomagnetic consistency test does not depend on the absolute rates of plate motion, but only on the geometry of the motion. To obtain adequate data, it is necessary to sample enough lava flows at one locality to average out the full range of geomagnetic secular variation. In addition, this locality must be at a point along the Hawaiian–Emperor chain sufficiently far north of Hawaii that, if the rocks were formed at their present latitude, their mean paleomagnetic inclination would differ significantly from the mean paleomagnetic inclination for the Brunhes epoch lavas of the island of Hawaii [5]. Hawaii is at 19.5°N, Midway Atoll is at 28.2°N , and the corresponding difference in magnetic inclination from the dipole formula is 12°. This difference is large enough to be detected paleomagnetically, especially if the low secular variation characteristic of the central Pacific basin for the past 700 000 years has been a permanent feature for the past several tens of millions of years [18, 19].

2. Paleomagnetism of basalts beneath Midway Atoll

In 1965, two deep holes were drilled on Midway Atoll through the reef limestones and into the underlying basalts [20]. The shorter of the two holes, called the Sand Island hole, was drilled to a depth of 173 m (568 ft.) and penetrated 19 m (61 ft.) of basalt at the bottom. The longer hole, called the Reef hole, was drilled to 505 m (1654 ft.) and penetrated 120 m (394 ft.) of basalt. Total core recovery from the basalt sections of both holes was 46%. A physical description of the basalt cores and a discussion of their petrology have been published by Macdonald [21]. That the lavas were erupted subaerially is indicated by lack of pillow structures, hyaloclastite, or tuff and by the presence of soil zones and reddened oxidized zones. The flows are both pahoehoe and aa, and the pahoehoe flows are moderately to highly vesicular; both types are extensively altered, mostly to clay minerals. Macdonald [21] concluded that because the nature of the alteration of the Midway lavas (except for the reddening and soil formation) is very different from the weathering and hydrothermal alteration of lavas of the Hawaiian Islands, it is the result of long submergence in sea water. From petrographic examination and from chemical analyses of the least altered parts of the cores, Macdonald [21] also concluded that the lavas are tholeiitic olivine basalt. Alkalic olivine basalts may be present in the upper part of the section, and mugearite pebbles were found in the overlying sedimentary rocks. Thus these cores very likely represent the end of

Table 1
Average paleomagnetic data from individual core segments.

Flow	Depth	N	H	R	k	δ	I	$J \pm$ s.d., $\times 10^3$
				Reef hole				
	1261	4	200	3.99552	669	3.1	8.5	4.3 ± 0.1
	1262	4	0	3.99836	1827	1.9	8.2	4.6 ± 0.5
R10	1265	3	0	2.99573	468	3.8	9.6	2.5 ± 0.6
	1269	4	0	3.99865	2218	1.7	4.5	5.8 ± 0.8
R 9	1271	4	0	3.99597	744	3.0	6.3	4.9 ± 0.9
	1272	4	0	3.99661	884	2.7	10.4	2.0 ± 0.1
R 8	1277	4	200	3.98967	290	4.8	41.4	1.7 ± 0.2
	1281	4	100	3.98894	271	4.9	39.1	5.7 ± 2.1
R 7	1288	4	50	3.99622	792	2.9	35.8	3.8 ± 1.2
	1292	3	0	2.99577	473	3.7	47.7	8.0 ± 1.0
R 6	1297	1	200	–	–	–	48.0	4.9 ±
	1302	4	0	3.99099	333	4.5	44.2	0.3 ± 0.03
	1309	3	200	2.94970	40	12.9*	46.0	0.3 ± 0.03
	1394	1	150	–	–	–	33.5	2.4
	1395	2	150	1.99827	578	2.4	25.0	4.3 ± 1.5
	1395	2	0	1.99677	309	3.3	36.4	12.9 ± 5.4
	1395	1	700	–	–	–	35.6	5.6
	1396	1	700	–	–	–	31.9	7.8
	1396	1	700	–	–	–	18.1	15.9
	1400	4	0	3.99805	1537	2.1	31.7	0.5 ± 0.03
	1401	2	200	1.99734	375	4.2	14.4	0.5 ± 0.1
R 5	1406	4	0	3.99907	3203	1.4	32.7	2.4 ± 0.3
	1409	4	0	3.99906	3162	1.5	23.7	11.0 ± 0.1
	1416	1	400	–	–	–	40.2	5.6
	1418	2	500	1.99989	9087	0.6	29.7	7.3 ± 0.04
	1419	1	500	–	–	–	27.8	6.7
	1419**	1	700	–	–	–	28.1	8.0
	1419**	3	150	2.99971	6972	0.8	31.3	9.5 ± 0.4
	1420	1	700	–	–	–	25.1	8.4
	1420**	1	700	–	–	–	22.4	16.6
	1591	4	0	3.98528	204	5.7	23.5	1.8 ± 0.9
R 4	1594	1	200	–	–	–	25.2	3.3
	1595	4	0	3.99465	561	3.4	15.3	2.7 ± 0.3
	1600	1	200	–	–	–	21.6	1.4
	1602	4	200	3.99704	1012	2.6	22.3	1.2 ± 0.04
	1606	4	0	3.99378	482	3.7	18.8	3.4 ± 0.3
	1611	2	500	1.99990	10 000	0.6	20.6	6.4 ± 0.2
	1611	2	700	1.99995	9334	0.4	22.0	4.9 ± 0.8
R 3	1612	4	0	3.99900	2991	1.3	26.1	3.6 ± 0.4
	1612	2	300	1.99995	8824	0.4	19.2	4.0 ± 0.08
	1613	5	300	4.99825	2283	1.5	23.6	3.4 ± 0.4
	1614**	6	700	5.99884	4299	1.1	22.0	4.9 ± 1.0
	1614**	2	500	1.99998	7867	0.2	24.2	7.6 ± 0.3
	1615**	3	500	2.99987	5629	0.5	27.5	5.1 ± 0.3
	1618	4	50	3.99114	338	4.4	23.1	5.7 ± 1.4
	1621	4	100	3.99758	1235	2.3	20.1	4.6 ± 0.2
R 2	1624	4	200	3.99256	403	4.0	18.5	1.0 ± 0.4
	1627	4	0	3.99336	452	3.8	23.0	0.7 ± 0.1

Table 1 (continued)

Flow	Depth	N	H	R	k	δ	I	$J \pm$ s.d., $\times 10^3$
	1632	4	0	3.99616	780	2.9	11.8	2.2 ± 0.07
	1638	4	0	3.99785	1393	2.2	33.4	2.2 ± 0.1
R 1	1642	4	0	3.99885	2606	1.6	52.1	1.7 ± 0.09
	1649	4	0	3.99862	2164	1.8	47.4	1.5 ± 0.06
				Sand Island hole				
S 3	527	4	0	3.99393	493	3.7	37.4	4.9 ± 0.4
	532	4	0	3.99523	628	3.2	28.2	5.7 ± 1.5
S 2	562	4	0	3.99866	2232	1.7	27.2	4.7 ± 0.2
	563	4	0	3.99848	1971	1.8	33.9	4.4 ± 0.2
S 1	567	3	400	2.99385	325	4.5	21.3	1.3 ± 0.2

Depth in feet, corresponding to core logs of Macdonald (1969).
N = number of specimens from each core segment.
H = peak alternating field used for demagnetization.
R = resultant of N unit vectors.
k = precision parameter (Fisher, 1953).
δ = root-mean-square angular dispersion of N directions.
I = mean inclination of remanent magnetization, degrees, positive downward.
J = intensity of remanent magnetization before demagnetization, emu/cm^3, with standard deviation (s.d.).
* Large angular dispersion due to shattered core.
** Fully oriented cores.

the main tholeiitic shield-building phase of the Midway volcano.

Potassium–argon ages have been determined by M.A. Lanphere from two essentially unaltered samples of basalt from depths of 486 m (1594 ft.) and 488 m (1600 ft.) in the Reef hole. The ages, reported by Ladd et al. [22], are 15.7 ± 0.9 and 16.6 ± 0.9 my, respectively. Because these radiometric ages appeared to be younger than the age indicated by Foraminifera in the overlying reef limestones, Ladd et al. [22] concluded that the rocks had lost radiogenic argon. Both McDougall [15] and Jackson et al. [17] have pointed out that revisions in the Tertiary faunal time scale have removed this discrepancy. Additional age determinations have recently been obtained from samples from the Reef hole. Including the above data, there are now seven measurements from four separate lava flows. The best weighted mean of these is 17.9 ± 0.6 my (M.A. Lanphere and G.B. Dalrymple, personal communication).

Magnetic measurements were made on a total of 57 separate core segments from both holes, including those used for the age determinations. The azimuthal orientations of most of the segments are unknown, but the upper ends of all segments had been marked so that the paleomagnetic polarities can be unambiguously determined. From one to six separate specimens were cut from each segment, and their remanent magnetizations (RM) were measured with a spinner magnetometer. Single specimens from each core segment were partially demagnetized in alternating fields up to 700 peak Oe. In only four of the core segments did the RM directions change greatly during demagnetization, and in most segments the directions did not change measurably. This indicates that little or no secondary magnetization was acquired by the rocks during drilling and further suggests that the weathering of the lavas did not greatly affect the stability of RM.

Magnetic data for all core segments are given in table 1 in the form of averages for all specimens cut from single segments. The statistics (R, k, δ) in table 1 thus reflect only the precision of relative orientation and measurements within single core segments. For roughly half the segments no magnetic cleaning was necessary, and the natural RM data are given. The inclinations of RM directions from table 1 are plotted in figs. 2 and 3 as a function of stratigraphic position

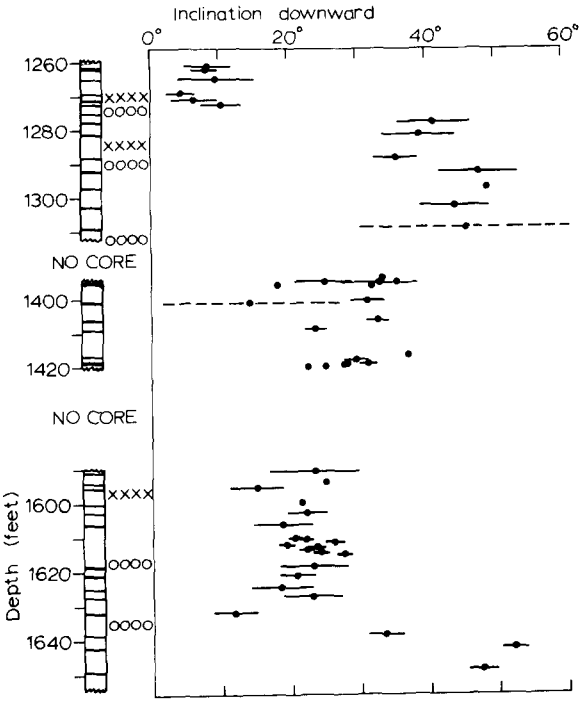


Fig. 2. Mean inclinations of remanent magnetization in core segments from Reef hole, Midway Atoll. Thicknesses of horizontal bars on column correspond to lengths of core segments. See text for explanation of confidence intervals on inclinations.

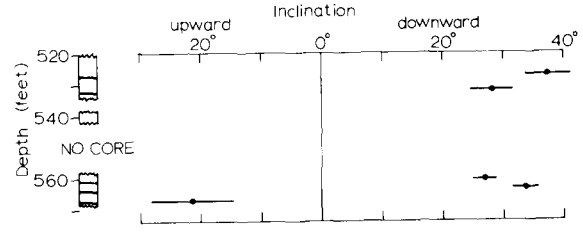


Fig. 3. Mean inclinations of remanent magnetization in core segments from Sand Island hole, Midway Atoll. See fig. 2 for explanation.

in the two drill holes. The confidence intervals for the inclinations are simply values of α_{95} calculated from N and R in table 1 and are equivalent to two standard deviations of the mean inclinations.

In order to decide whether these measurements cover enough time to sample the full range of secular variation and to calculate the mean paleomagnetic inclination and its uncertainty correctly, it is necessary to know how many separate lava flows have been sampled. Because of incomplete core recovery, this determination is not straightforward, and the criteria for identifying flow boundaries adopted are: (1) red (oxidized) zones, with or without soil development, indicating protracted subaerial weathering; (2) clinkery zones, interpreted as the autobrecciated tops of aa flows; (3) long uncored intervals; and (4) reversal of inclination of RM direction. The occurrences of these four criteria are shown in figs. 2 and 3, and the grouping of core segments into single lava flows is indicated in table 1.

The total number of flows sampled is apparently 13. Only one reversal was found, at the bottom of the Sand Island hole. There is a definite tendency for the RM inclinations to change less across a flow boundary identified by a clinkery zone than across other flow boundaries; this may be expected because autobrecciated zones may occur within flows or between flow units of a single eruption and hence represent little or no time lapse. Omission of this criterion would only reduce the total number of flows from 13 to 10, however, and large changes in inclination occur where there is no indication of a flow boundary, such as between 1400 and 1416 ft. and between 1638 and 1642 ft. in the Reef hole (fig. 2). The cause of these variations is unknown because the smooth and straight appearance of the cores rules out

Table 2
Flow-average paleomagnetic data (means of core segment means).

Flow	N	I	s.d	VGL	s.d
R10	4	7.7	2.2	3.9	1.1
R 9	2	8.4	1.2	4.2	0.6
R 8	2	40.3	1.6	23.0	1.2
R 7	1	35.8	–	19.8	–
R 6	4	46.5	1.8	27.8	1.5
R 5	17	28.5	6.5	15.2	3.9
R 4	3	21.3	5.3	11.0	2.9
R 3	11	22.5	2.7	11.7	1.5
R 2	5	19.3	4.6	9.9	2.5
R 1	3	44.3	9.7	26.0	7.6
S 3	2	32.8	6.5	17.9	4.2
S 2	2	30.6	4.7	16.5	2.9
S 1	1	−21.3	–	11.0	–
			2σ		2σ
Mean	13	27.6	12.4 ± 6.8	14.7	7.6 ± 4.2

Symbols as in table 1. s.d. = standard deviation of data. 2σ= two standard deviations of mean. VGL = virtual geomagnetic latitude (see text).

short-wavelength crookedness of the drill holes. Hence, 13 is a reasonable estimate of the number of flows sampled.

Average paleomagnetic inclinations for the 13 flows are given in table 2; these are calculated from the core-segment means in table 1. Also in table 2 are the virtual latitudes found by applying the dipole formula to the flow-mean inclinations. These virtual geomagnetic latitudes, a term introduced for convenience here, have the same significance as the virtual geomagnetic poles (VGP's) calculated from conventional paleomagnetic determinations on single lava flows [24]. The standard deviations are found by multiplying the standard deviations of the inclinations by $\frac{1}{2}(1 + 3\sin^2 L)$ where L is the virtual latitude [24]. The mean inclination and latitude in table 2 are the averages of the mean inclinations and virtual latitudes given in the table.

The dispersion of the data in table 2 is rather great, and two possible causes are deviations of the drill holes from the vertical and geomagnetic secular variation. Only one point in the drill holes was surveyed, at 451 m in the Reef hole where the deviation is 15m [20]. This is equivalent to an angular deflection of only 2° which, together with the straightness of the core segments, implies that deflections of the drill hole do not contribute materially to the 12.4° dispersion of the mean inclinations nor is the overall mean inclination significantly affected.

The other cause of dispersion, secular variation, may be evaluated as follows: Extensive paleomagnetic data are available for eight major series of lavas of the Hawaiian Islands, ranging in mean age from 5000 years to 4.5 my [5–7, 19, 25]. Standard deviations of virtual paleomagnetic latitudes of the Hawaiian Islands have been calculated from the VGP positions given in these sources; the standard deviations range from 1.4° for the 5000-year-old Kau Volcanic Series of Hawaii [6] to 21.2° for the 3.1 my-old Waianae Volcanic Series of Oahu [19]. The value of 7.6° for the Midway lavas (table 2) is similar to the value of 7.5° calculated for 197 Brunhes-age lavas of the island of Hawaii, including the Kau Volcanic Series [5, 6]. It appears from this comparison that the dispersion of inclinations in the Midway lavas is due to secular variation, and that enough lavas representing sufficient time have been measured to cover much of the probable full range of secular variation.

During drilling of the Reef hole, three attempts were made to obtain azimuthally oriented cores, starting at depths of 1393, 1418, and 1611 ft. These cores

Table 3
A.F. demagnetization results and overall mean RM directions in fully oriented core segments from flows R5 and R3.

H	D	I	R	k	α_{95}
Flow R5: 5 specimens from 3 core segments (N = 3)					
0	357.3	30.4	2.98566	139	10.5
150	358.5	29.6	2.98444	129	10.9
300	358.9	29.4	2.98327	120	11.3
500	359.5	28.1	2.98761	161	9.7
700	358.8	26.6	2.99174	242	7.9
Flow R3: 11 specimens from 3 core segments (N = 3)					
0	356.4	26.6	2.99655	580	5.1
150	356.9	25.5	2.99473	379	6.3
300	356.9	25.1	2.99585	482	5.6
500	356.4	24.9	2.99585	482	5.6
700	356.3	24.3	2.99640	555	5.2
Both flows: 16 specimens from 6 segments (N = 2)					
0	356.8	28.5	1.99888	892	8.4
700	357.5	25.5	1.99923	1302	6.9

D = declination east of north.
N = number of samples used in calculating k and α_{95}.
α_{95} = semiangle of 95% confidence cone [23].
Other symbols as in table 1.

were taken with a nonmagnetic core barrel having a compass and camera in its upper part and lugs on its lower end that inscribe the core as it enters the tube. Rotation of the drill string and fluid pumping were stopped each time the compass was photographed. Because of the decomposed nature of the basalt, no orientation marks were visible on the core from 1393 ft., but the orientation procedure was successful for the other two attempts. Three oriented segments were obtained from flow R5 between 1419 and 1420 ft., and three were obtained from flow R3 between 1614 and 1615 ft. These segments are identified in table 1.

Complete A.F. demagnetization experiments were done on all specimens obtained from these oriented core segments, and the results are summarized in table 3. For each of those segments (all but two) from which more than one specimen were obtained, the specimen RM directions were averaged prior to calculating the segments averages and statistics in table 3. The RM directions were stable to demagnetizing fields of 700 Oe and did not change significantly during demagnetization. Trial demagnetizations at higher fields resulted in greatly increased dispersion of RM directions.

Although flows R5 and R3 are separated by a vertical distance of about 55 m, no core was recovered from much of this interval and no oxidized zones were observed in the core that was recovered from it. Hence the fact that the RM directions in flows R5 and R3 are not significantly different may mean that these flows and those intervening were erupted within a short interval of time. For this reason, the pole position obtained from the data in table 3 that was quoted by Francheteau et al. [2] is only a virtual magnetic pole in the sense of Cox and Doell [24] and is not a true paleomagnetic pole.

3. Discussion and conclusions

Presently available indications of the ancient latitudes of the Pacific plate are shown in fig. 4, in which the location of Midway Atoll is used as a reference point on the plate. This diagram compares the latitudinal movement predicted by the Wilson-Morgan hypothesis of fixed melting spots with that indicated by the paleomagnetic data. The paleolatitudes predicted by the fixed-melting-spot hypothesis are shown by a dashed line, which is constructed in the following way: The point at age = 0 is the present of Midway Atoll. The point at age = 18 my is obtained by moving the Midway reference point to the present latitude of the volcanically active island of Hawaii, the presumed permanent location of the melting spot and hence the place where nearly all of the Midway volcano was formed 18 my ago. Assuming constant plate motion during the time of generation of a single island chain, this line may be extended to the point representing the difference in latitude between Hawaii and the juncture of the Hawaiian and Emperor chains

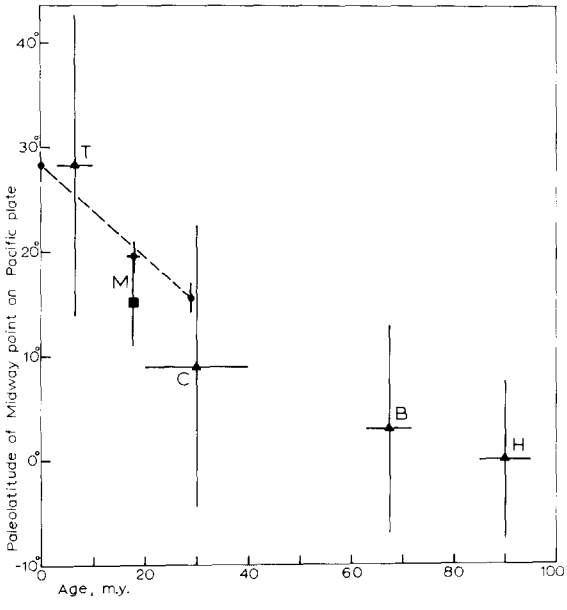


Fig. 4. Paleolatitude of the Pacific plate at the point represented by Midway Atoll. Heavy dashed line is the paleolatitude predicted by assuming all volcanoes of the Hawaiian chain were formed at the latitude of Hawaii [12] and using the available radiometric age data summarized by Jackson et al. [17]. Points (circles) are present location of Midway at age = 0, Midway (M) at 17.9 my, and Hawaiian–Emperor bend at 29 my. Vertical bars on these points represent the estimated 300-km width of the melting spot [17], and horizontal bar represents estimated error in radiometric ages of Midway. The paleomagnetic latitude of Midway (table 2) is indicated by a square (M). The triangle marked B is the estimated latitude of formation of the Great Magnetic Bight in anomalies 25–32 [3]. The remaining three points are obtained from the pole positions calculated by Francheteau et al. [2]: T, Tripod seamounts: C, California seamounts; H, Hawaiian seamounts (see text for a discussion of the uncertainties in these).

(fig. 1), i.e., by moving the "bend" to the latitude of Hawaii. The extrapolated age for this point using published and unpublished radiometric ages from Midway and the Hawaiian Islands is 29 my (G.B. Dalrymple, personal communication).

Paleomagnetically determined ancient latitudes for the Pacific plate are shown in fig. 4. For the three values obtained from seamount surveys (Tripod, California, and Hawaiian) the paleolatitudes are found by measuring the great-circle distances to Midway from the paleomagnetic poles given by Francheteau et al. [2]. The associated paleolatitude uncertainties, which are large, are simply the values of α_{95} given by those authors for the paleomagnetic poles. The age uncertainties are rough estimates following Francheteau et al. [2]: the maximum ages of the Tripod and California seamounts are fixed by the ages of the magnetic anomalies on which they lie, and both fossil and radiometric ages are available for the Hawaiian seamounts, as summarized by those authors. The age range for the Great Magnetic Bight is the ages of anomalies 25–32 that form it [26] and an estimate for latitude uncertainty of $\pm 10^\circ$ is arbitrarily assigned here. Note that the uncertainties in paleolatitudes are far greater than the uncertainties in ages of these features. The uncertainty in the paleomagnetically determined latitude of Midway is that given in table 2, and the age range of $\pm$ 1.3 my is based on analytical uncertainty only [17]. No paleolatitude is shown in fig. 4 for the paleomagnetic pole from the Japanese seamounts because they and the Hawaiian seamounts may have undergone relative displacements since their formation [2].

Fig. 4 shows clearly that the latitudinal motion of the Pacific plate predicted by the fixed-melting-spot hypothesis is broadly similar to the motion indicated by the paleomagnetic results. This diagram also shows the lack of precision of the presently available paleomagnetic data for the Pacific plate. The use of paleomagnetic poles to infer ancient latitudes depends on the well-known but often unstated assumption that the average geomagnetic field has been that of a geocentric axial dipole over the time span represented by the rocks sampled. Put another way, in order to use paleomagnetic poles to test the Wilson-Morgan hypothesis of fixed melting spots, true polar wandering must not have occurred. According to Morgan [12], the absolute motion of the North American plate is very small and has evidently been small since the age of the Hawaiian–Emperor bend. An extensive paleomagnetic study of the Early to Middle Miocene lavas of the Columbia Plateaus has been published by Watkins [27]: the paleomagnetic pole obtained from 433 lavas does not differ significantly from the geographic pole. Hence within the limits of resolution of the paleomagnetic method, true polar wandering has apparently not occurred since the time of eruption of the Midway lavas.

The paleolatitude that would be predicted for Midway from McDougall's [15] hypothesis of a southward-moving volcanic source is halfway between its present latitude and that of Hawaii, or 23.8°. The paleomagnetic latitude of Midway, $14.7^\circ \pm 4.2^\circ$ (table 2) is significantly south of this and is not significantly different from the probable present latitude of the melting spot beneath Hawaii (figs. 1 and 4). Hence the paleomagnetic data from the Midway Atoll lavas, in addition to indicating a 1400-km northward movement of Midway in the last 18 my, appear to rule out McDougall's hypothesis and to support the hypothesis that the volcanic source or melting spot beneath Hawaii has remained fixed relative to the mantle below the asthenosphere during the time of formation of the Hawaiian volcanic island chain. If on the other hand the paleomagnetically determined paleolatitude of Midway is interpreted literally without considering the uncertainties shown in fig. 4, and no true polar wandering occurred, a northward movement of the Hawaii melting spot of 5° during the last 18 my is indicated.

In obtaining their preliminary polar wandering curve for the Pacific plate, Francheteau et al. [2] used only the paleomagnetic pole from the Hawaiian seamounts, the virtual pole from the two Midway lavas, and the present geographic pole. They did not use the paleomagnetic pole from the California seamounts because of the large uncertainty in the age of these seamounts. We have indicated above that the Midway pole does not include the full range of geomagnetic secular variation and hence cannot be used to help define a polar wandering curve. At the present time, there are not sufficient paleomagnetic data from the Pacific plate to construct a polar path. According to the fixed-melting-spot hypothesis, the abrupt bend in the Hawaiian-Emperor chain (fig. 1) marks a major change in the direction of absolute motion of the Pacific plate. By extrapolating from the known ages along the Hawaiian chain, the age of the bend has been variously estimated as 40–55 my [12] and more recently as 25 my [17] and

29 my (G.B. Dalrymple, personal communication). From an analysis of the magnetic anomalies along the Eltanin fracture zone in the southeast Pacific, however, Herron [28] has concluded that the major change in Pacific plate motion could have occurred only prior to 80 my ago. In fig. 4 there is a suggestion of a change in the rate of latitudinal motion at about 30 my. Clearly the significance of the bends in the aseismic ridges shown in fig. 1 is still not well established, but may become so when enough paleomagnetic information becomes available for a detailed polar wandering path to be constructed for the Pacific plate.

Acknowledgements

We wish to thank Professor G.A. Macdonald and Dr. Harry S. Ladd for making the Midway cores available, and M. A. Lanphere, E.A. Silver, and G.B. Dalrymple for helpful discussions. Many of the magnetic measurements were made by E.A. Mankinen and Mrs. Naoma Dorety.

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